$$+ O_\infty(\tilde{\eta}_V \mathbb{E}[T^{-1}] \cdot N_{\mathrm{trg}}/N).$$

B.2.2 Detailed Calculations

Let

$$\begin{aligned}
&\boldsymbol{\Delta}(k,T) \\
:=&\frac{1}{T}\sum_{t=1}^{T}\left(\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_T^\top\right] - \mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_T^\top | z_{T+1} = k\right]\right)
\end{aligned}$$

for $N_{\mathrm{trg}} + 1 \leq k \leq N$. Then, it holds that $\nabla_{\boldsymbol{W}_{KQ}} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V^*) = \frac{1}{N - N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \mathbb{E}_T[\boldsymbol{\Delta}(k,T)] + \tilde{\eta_V} \mathbb{E}[T^{-1}] O_\infty(N_{\mathrm{trg}}/N)$.

For the first and second (block) columns, we have the following lemma:

Lemma 11. *Let $\boldsymbol{W}_{KQ}^* = -\eta_{KQ} \nabla_{\boldsymbol{W}_{KQ}} \bar{\mathcal{L}}(\boldsymbol{W}_{KQ}, \boldsymbol{W}_V^*)$. Then, it holds that*

$$\begin{aligned}
&\boldsymbol{W}_{KQ}^*[:, 1 : L+N] \\
=&\tilde{\eta_V} \eta_{KQ} \frac{1}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} \sum_{\ell} p(\ell) \Bigg(T(\ell)^{-1} \begin{bmatrix} \boldsymbol{p}_{\ell+2} \mathbb{E}_{\ell'}[T(\ell')^{-1}] + \boldsymbol{p}_{\ell+3} \mathbb{E}_{\ell'}[T(\ell')^{-1}] \\ \boldsymbol{0} \\ \boldsymbol{e}_w \mathbb{E}_{\ell'}[T(\ell')^{-1}] \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_{T(\ell)}^\top & \boldsymbol{e}_w^\top \end{bmatrix} \\
&- 2T(\ell)^{-2} \begin{bmatrix} \boldsymbol{1}_{1:T(\ell)\setminus\{\ell+2,\ell+3\}} \mathbb{E}_{\ell'}[T(\ell')^{-1}] \\ 2\boldsymbol{e}_w \mathbb{E}_{\ell'}[T(\ell')^{-1}] \\ \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_{T(\ell)}^\top & \boldsymbol{e}_w^\top \end{bmatrix} \Bigg) \\
&+ \tilde{\eta_V} \eta_{KQ} \mathbb{E}_{\ell'}[T(\ell')^{-1}] O(N_{\mathrm{trg}} \cdot N^{-1}).
\end{aligned}$$

Here $O(N_{\mathrm{trg}} \cdot N^{-1})$ denotes a matrix whose entries are all of $O(N_{\mathrm{trg}} \cdot N^{-1})$ and $p(\ell)$ is the probability of drawing ℓ at pretraining data.

Proof. [Proof of Lemma 11] Suppose $\boldsymbol{W}_V$ is obtained by Lemma 9. Note that

$$\begin{aligned}
&\boldsymbol{\Delta}(k,T)[:, : L] \\
&= \frac{1}{T}\sum_{t=1}^{T}\left(\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{e}_k^\top \boldsymbol{W}_V \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\right] - \mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{e}_k^\top \boldsymbol{W}_V \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) | z_{T+1} = k\right]\right) \boldsymbol{p}_T^\top \\
&= \frac{1}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} \frac{1}{T} \sum_{t=1}^{T}\left(\mathbb{E}_{\boldsymbol{X}}\left[(\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_t^\top | \boldsymbol{x}_T = w\right] - \mathbb{E}_{\boldsymbol{X}}\left[(\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_t^\top | z_{T+1} = k, \boldsymbol{x}_T = w\right]\right) \boldsymbol{W}_V{}^\top \boldsymbol{e}_k \boldsymbol{p}_T^\top
\end{aligned}$$

and similarly

$$\begin{aligned}
&\boldsymbol{\Delta}(k,T)[:, L+1 : L+N] \\
=&\frac{1}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} \left[\frac{1}{T} \sum_{t=1}^{T}\left(\mathbb{E}_{\boldsymbol{X}}\left[(\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_t^\top | \boldsymbol{x}_T = w\right] - \mathbb{E}_{\boldsymbol{X}}\left[(\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_t^\top | z_{T+1} = k, \boldsymbol{x}_T = w\right]\right) \cdot \boldsymbol{W}_V{}^\top \boldsymbol{e}_k \boldsymbol{e}_w^\top \right].
\end{aligned}$$

Now let us consider the difference

$$\begin{aligned}
&\frac{1}{T} \sum_{t=1}^{T}\left(\mathbb{E}_{\boldsymbol{X}}\left[(\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_t^\top | \boldsymbol{x}_T = w\right] - \mathbb{E}_{\boldsymbol{X}}\left[(\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T})\boldsymbol{x}_t^\top | z_{T+1} = k, \boldsymbol{x}_T = w\right]\right) \\
=&\frac{1}{T} \sum_{t=1}^{T}\left(\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t \boldsymbol{x}_t^\top | \boldsymbol{x}_T = w\right] - \mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t \boldsymbol{x}_t^\top | z_{T+1} = k, \boldsymbol{x}_T = w\right]\right)
\end{aligned}$$

$$
\begin{aligned}
&- \frac{1}{T^2}\sum_{t=1}^{T}\sum_{t'=1}^{T}\big(\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t\boldsymbol{x}_{t'}^\top|\boldsymbol{x}_T=w\right]-\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t\boldsymbol{x}_{t'}^\top|z_{T+1}=k,\boldsymbol{x}_T=w\right]\big)\\
=&\left(\frac{1}{T}-\frac{1}{T^2}\right)\sum_{t=1}^{T}\big(\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t\boldsymbol{x}_t^\top\big|\boldsymbol{x}_T=w\right]-\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t\boldsymbol{x}_t^\top|z_{T+1}=k,\boldsymbol{x}_T=w\right]\big)\\
&- \frac{1}{T^2}\sum_{t\neq t'}\big(\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t\boldsymbol{x}_{t'}^\top|\boldsymbol{x}_T=w\right]-\mathbb{E}_{\boldsymbol{X}}\left[\boldsymbol{x}_t\boldsymbol{x}_{t'}^\top|z_{T+1}=k,\boldsymbol{x}_T=w\right]\big)
\end{aligned}
$$

Then it suffices to calculate the vector

$$
\boldsymbol{d}(t,t',k,w)=\boldsymbol{M}(t,t',k,w)\boldsymbol{W}_V^\top\boldsymbol{e}_k\coloneqq\big(\mathbb{E}[\boldsymbol{x}_t\boldsymbol{x}_{t'}^\top|\boldsymbol{x}_T=w]-\mathbb{E}[\boldsymbol{x}_t\boldsymbol{x}_{t'}^\top|z_{T+1}=k,\boldsymbol{x}_T=w]\big)\boldsymbol{W}_V^\top\boldsymbol{e}_k
$$

for each t,t',k and w. Recall

$$
\boldsymbol{W}_V^\top\boldsymbol{e}_k=\tilde{\eta_V}\left[\begin{array}{c}
\frac{\alpha_1 N_{\mathrm{trg}}}{N-N_{\mathrm{trg}}}\\
\vdots\\
\frac{\alpha_L N_{\mathrm{trg}}}{N-N_{\mathrm{trg}}}\\
\hline
\frac{2}{N-N_{\mathrm{trg}}}\mathbb{E}[T^{-1}]\mathbf{1}_{N_{\mathrm{trg}}}\\
(-\frac{1}{N}\mathbb{E}[T^{-1}]+O(N_{\mathrm{trg}}\cdot N^{-2}))\mathbf{1}_{k-N_{\mathrm{trg}}-1}\\
\mathbb{E}[T^{-1}]+\mathbb{E}[T^{-1}]O(N_{\mathrm{trg}}\cdot N^{-1})+O(N_{\mathrm{trg}}\cdot N^{-2})\\
(-\frac{1}{N}\mathbb{E}[T^{-1}]+O(N_{\mathrm{trg}}\cdot N^{-2}))\mathbf{1}_{N-k}\\
\hline
\frac{1}{N-N_{\mathrm{trg}}}\mathbb{E}[T^{-1}]\mathbf{1}_{N_{\mathrm{trg}}}\\
(-\frac{1}{N}\mathbb{E}[T^{-1}]+O(N_{\mathrm{trg}}\cdot N^{-2}))\mathbf{1}_{k-N_{\mathrm{trg}}-1}\\
\mathbb{E}[T^{-1}]+\mathbb{E}[T^{-1}]O(N_{\mathrm{trg}}\cdot N^{-1})+O(N_{\mathrm{trg}}\cdot N^{-2})\\
(-\frac{1}{N}\mathbb{E}[T^{-1}]+O(N_{\mathrm{trg}}\cdot N^{-2}))\mathbf{1}_{N-k}
\end{array}\right].
$$

For preparation, we define some vectors: let

$$
\boldsymbol{\alpha}(k)=\left[\underbrace{0,0,\ldots,0}_{N_{\mathrm{trg}}\text{ zeros}},\frac{1}{N-N_{\mathrm{trg}}},\ldots,\frac{1}{N-N_{\mathrm{trg}}},\underbrace{\frac{N+N_{\mathrm{trg}}+1}{N-N_{\mathrm{trg}}}}_{k\text{-th entry}},\frac{1}{N-N_{\mathrm{trg}}},\ldots,\frac{1}{N-N_{\mathrm{trg}}}\right]^\top\in\mathbb{R}^N
$$

for $N_{\mathrm{trg}}+1\leq k\leq N$ and

$$
\boldsymbol{\beta}=\left[\underbrace{0,0,\ldots,0}_{N_{\mathrm{trg}}\text{ zeros}},\frac{1}{N-N_{\mathrm{trg}}},\ldots,\frac{1}{N-N_{\mathrm{trg}}}\right]^\top\in\mathbb{R}^N.
$$

First, if $t=t'$, then $\boldsymbol{M}(t,t',k,w)$ is zero unless $t=\ell+2$ or $t=\ell+3$, as z_{T+1} is independent of z_i ($i\in[T],i\neq\ell+2$) and only $\boldsymbol{x}_{\ell+2}$ and $\boldsymbol{x}_{\ell+3}$ include the information of $z_{\ell+2}$. For each case, we have

$$
\boldsymbol{M}(\ell+2,\ell+2,k,w)=\begin{bmatrix}\boldsymbol{O}_{L\times L} & \boldsymbol{p}_{\ell+2}\boldsymbol{\alpha}(k)^\top & \boldsymbol{O}_{L\times N}\\ \boldsymbol{\alpha}(k)\boldsymbol{p}_{\ell+2}^\top & \mathrm{diag}(\boldsymbol{\alpha}(k)) & \boldsymbol{\alpha}(k)\boldsymbol{e}_w^\top\\ \boldsymbol{O}_{N\times L} & \boldsymbol{e}_w\boldsymbol{\alpha}(k)^\top & \boldsymbol{O}_{N\times N}\end{bmatrix}
$$

and

$$
\boldsymbol{M}(\ell+3,\ell+3,k,w)=\begin{bmatrix}\boldsymbol{O}_{L\times L} & \boldsymbol{O}_{L\times N} & \boldsymbol{p}_{\ell+3}\boldsymbol{\alpha}(k)^\top\\ \boldsymbol{O}_{N\times L} & \boldsymbol{O}_{N\times N} & \boldsymbol{\beta}\boldsymbol{\alpha}(k)^\top\\ \boldsymbol{\alpha}(k)\boldsymbol{p}_{\ell+3}^\top & \boldsymbol{\alpha}(k)\boldsymbol{\beta}^\top & \mathrm{diag}(\boldsymbol{\alpha}(k))\end{bmatrix},
$$

then we obtain

$$\boldsymbol{d}(\ell+2,\ell+2,k,w)=\tilde{\eta_V}\begin{bmatrix}-\boldsymbol{p}_{\ell+2}\mathbb{E}[T^{-1}]\\-\boldsymbol{e}_k\mathbb{E}[T^{-1}]\\-\boldsymbol{e}_w\mathbb{E}[T^{-1}]\end{bmatrix}+\tilde{\eta_V}O_\infty(\mathbb{E}[T^{-1}]N_{\mathrm{trg}}\cdot N^{-1}+N_{\mathrm{trg}}\cdot N^{-2})$$

and

$$\boldsymbol{d}(\ell+3,\ell+3,k,w)=\tilde{\eta_V}\begin{bmatrix}-\boldsymbol{p}_{\ell+3}\mathbb{E}[T^{-1}]\\\boldsymbol{0}\\-\boldsymbol{e}_k\mathbb{E}[T^{-1}]\end{bmatrix}+\tilde{\eta_V}O_\infty(\mathbb{E}[T^{-1}]N_{\mathrm{trg}}\cdot N^{-1}+N_{\mathrm{trg}}\cdot N^{-2}).$$

For the case $t\neq t'$, deal with the following three cases:

(i) If $(t,t')=(\ell+2,\ell+3)$ or $(t,t')=(\ell+3,\ell+2)$ then we have

$$\begin{aligned}&\boldsymbol{M}(\ell+2,\ell+3,k,w)+\boldsymbol{M}(\ell+3,\ell+2,k,w)\\=&\begin{bmatrix}\boldsymbol{O}_{L\times L}&\boldsymbol{p}_{\ell+3}\boldsymbol{\alpha}(k)^\top&\boldsymbol{p}_{\ell+2}\boldsymbol{\alpha}(k)^\top\\\boldsymbol{\alpha}(k)\boldsymbol{p}_{\ell+3}^\top&\boldsymbol{\alpha}(k)\boldsymbol{\beta}^\top+\boldsymbol{\beta}\boldsymbol{\alpha}(k)^\top&\mathrm{diag}(\boldsymbol{\alpha}(k))\\\boldsymbol{\alpha}(k)\boldsymbol{p}_{\ell+2}^\top&\mathrm{diag}(\boldsymbol{\alpha}(k))&\boldsymbol{\alpha}(k)\boldsymbol{e}_w^\top+\boldsymbol{e}_w\boldsymbol{\alpha}(k)^\top\end{bmatrix}\end{aligned}$$

and

$$\begin{aligned}&\boldsymbol{d}(\ell+2,\ell+3,k,w)+\boldsymbol{d}(\ell+3,\ell+2,k,w)\\=&\tilde{\eta_V}\begin{bmatrix}-\boldsymbol{p}_{\ell+2}\mathbb{E}[T^{-1}]-\boldsymbol{p}_{\ell+3}\mathbb{E}[T^{-1}]\\-\boldsymbol{e}_k\mathbb{E}[T^{-1}]\\-\boldsymbol{e}_k\mathbb{E}[T^{-1}]-\boldsymbol{e}_w\mathbb{E}[T^{-1}]\end{bmatrix}+\tilde{\eta_V}O_\infty(\mathbb{E}[T^{-1}]N_{\mathrm{trg}}\cdot N^{-1}+N_{\mathrm{trg}}\cdot N^{-2}).\end{aligned}$$

(ii) For $t\neq\ell+2,\ell+3$ we have

$$\boldsymbol{M}(t,\ell+2,k,w)+\boldsymbol{M}(\ell+2,t,k,w)=\begin{bmatrix}\boldsymbol{O}&\boldsymbol{p}_t\boldsymbol{\alpha}(k)^\top&\boldsymbol{O}\\\boldsymbol{\alpha}(k)\boldsymbol{p}_t^\top&\boldsymbol{\gamma}(t)\boldsymbol{\alpha}(k)^\top+\boldsymbol{\alpha}(k)\boldsymbol{\gamma}(t)^\top&\boldsymbol{\alpha}(k)\boldsymbol{\beta}^\top\\\boldsymbol{O}&\boldsymbol{\beta}\boldsymbol{\alpha}(k)^\top&\boldsymbol{O}\end{bmatrix}$$

where $\boldsymbol{\gamma}(t)=\boldsymbol{e}_w$ if $t=\ell+1$ or $t=T$ and $\boldsymbol{\gamma}(t)=\boldsymbol{\beta}$ otherwise. To summarize,

$$\boldsymbol{d}(t,\ell+2,k,w)+\boldsymbol{d}(\ell+2,t,k,w)=\tilde{\eta_V}\begin{bmatrix}-\boldsymbol{p}_t\mathbb{E}[T^{-1}]\\-\boldsymbol{e}_w\mathbb{E}[T^{-1}]\\\boldsymbol{0}\end{bmatrix}+\tilde{\eta_V}O_\infty(\mathbb{E}[T^{-1}]N_{\mathrm{trg}}\cdot N^{-1}+N_{\mathrm{trg}}\cdot N^{-2}).$$

if $t=\ell+1$ or $t=T$ and

$$\boldsymbol{d}(t,\ell+2,k,w)+\boldsymbol{d}(\ell+2,t,k,w)=\tilde{\eta_V}\begin{bmatrix}-\boldsymbol{p}_t\mathbb{E}[T^{-1}]\\\boldsymbol{0}\\\boldsymbol{0}\end{bmatrix}+\tilde{\eta_V}O_\infty(\mathbb{E}[T^{-1}]N_{\mathrm{trg}}\cdot N^{-1}+N_{\mathrm{trg}}\cdot N^{-2}).$$

otherwise.

(iii) For $t\neq\ell+2,\ell+3$ we have

$$\boldsymbol{M}(t,\ell+3,k,w)+\boldsymbol{M}(\ell+3,t,k,w)=\begin{bmatrix}\boldsymbol{O}&\boldsymbol{O}&\boldsymbol{p}_t\boldsymbol{\alpha}(k)^\top\\\boldsymbol{O}&\boldsymbol{O}&\boldsymbol{\gamma}(t)\boldsymbol{\alpha}(k)^\top\\\boldsymbol{\alpha}(k)\boldsymbol{p}_t^\top&\boldsymbol{\alpha}(k)\boldsymbol{\gamma}(t)^\top&\boldsymbol{\beta}\boldsymbol{\alpha}(k)^\top+\boldsymbol{\alpha}(k)\boldsymbol{\beta}^\top\end{bmatrix}$$

and we obtain

$$\boldsymbol{d}(t,\ell+3,k,w)+\boldsymbol{d}(\ell+3,t,k,w)=\tilde{\eta_V}\begin{bmatrix}-\boldsymbol{p}_t\mathbb{E}[T^{-1}]\\-\boldsymbol{e}_w\mathbb{E}[T^{-1}]\\\boldsymbol{0}\end{bmatrix}+\tilde{\eta_V}O_\infty(\mathbb{E}[T^{-1}]N_{\mathrm{trg}}\cdot N^{-1}+N_{\mathrm{trg}}\cdot N^{-2})$$